

LESSON 9-1 • HOMEWORK PACKET • SHEET 1 OF 2

Independent and Dependent Variables

Maryland Standard 6.EE.C.9 • Learning target – I can name the independent and dependent variables in a relationship.

REMEMBER

- The **independent variable** is the input you choose or control.
- The **dependent variable** is the output. It **depends** on the input.
- On a graph: independent on the **x-axis**, dependent on the **y-axis**.

WORKED EXAMPLE

The more cars Andre washes, the more money he earns. Name the variables.

Step 1 Andre controls how many cars he washes.

Step 2 Test: "Money earned depends on cars washed." It makes sense.

Answer Independent: cars washed • Dependent: money earned

PRACTICE*show your work • label units*

1. For each pair of quantities, write **I** under the independent variable and **D** under the dependent variable.

	Quantity	I or D	Quantity	I or D
a.	minutes biking		miles traveled	
b.	total cost		tickets bought	
c.	guests at a party		cups of juice needed	

2. **Bus Ride.** The longer Imani rides the bus, the farther she gets from home. Write a sentence about these two quantities that uses "depends on."

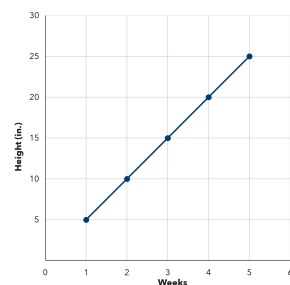
Sentence _____

3. **Dog Walking.** Rosa earns \$6 per dog she walks. Name both variables.

Independent / Dependent _____

4. **Sunflower.** Kenji graphs the height of his sunflower each week.

- a. Name the independent variable. _____
- b. Name the dependent variable. _____
- c. What does the point (4, 20) mean? _____



5. **Find the Error.** The number of people at a picnic changes how many sandwiches are needed. Leah says the sandwiches are independent because that is the bigger number. Explain her mistake.

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____

AT HOME Name two things at home that change together. Which one depends on the other?

LESSON 9-1 • HOMEWORK PACKET • SHEET 2 OF 2**Tables of Values**

Maryland Standard 6.EE.C.9 • Learning target – I can build a table of values for a relationship between two variables.

REMEMBER

- A **table of values** pairs each input (x) with its output (y).
- Use the **rule** on every input. With a fee, **multiply first**, then add.

WORKED EXAMPLE

A copier prints 20 pages each minute. Make a table for 1, 2, and 4 minutes.

Step 1 Minutes is x . Pages is y . The rule is $y = 20x$.**Step 2** $20 \times 1 = 20$, $20 \times 2 = 40$, $20 \times 4 = 80$ **Answer 20, 40, and 80 pages****PRACTICE***show your work • label units*

1. Use the rule
- $y = x + 8$
- to complete the table.

x	0	2	4	7
y				

- 2.
- Skate Rental.**
- A rink charges \$5 to get in plus \$3 for each hour of skating. Complete the table.

Hours (x)	1	2	3	6
Cost in \$ (y)				

- 3.
- Marker Boxes.**
- The table shows markers in boxes. Describe the rule. Then fill in the missing values.

Boxes (x)	1	2	3	4	8
Markers (y)	8	16	24		

Rule _____

- 4.
- Find the Error.**
- Marcus made a table for
- $y = 2x + 1$
- . He wrote
- x
- : 1, 2, 3, 4 and
- y
- : 3, 5, 7, 8. Which
- y
- value is wrong? What should it be?

Answer _____

- 5.
- Stretch.**
- A taxi charges \$4 plus \$2 for each mile. Complete the table. How many miles cost \$20?

Miles (x)	1	3	5	
Cost in \$ (y)				20

TEST PRACTICE*choose the best answer*

6. A rule is "multiply by 4, then add 5." What is
- y
- when
- $x = 6$
- ?

A 24**B** 44**C** 29**D** 15

7. A table has
- x
- values 1, 2, and 3. Which
- y
- values match the rule
- $y = x + 6$
- ?

A 6, 12, 18**B** 7, 8, 9**C** 7, 14, 21**D** 6, 7, 8**CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about # ____